

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: S.Y. B. Tech Branch: Electronics and Computer Engineering/ Semester: III

Electronics and Computer Science:

Subject Code & Name: (BTEPC303) Programming, Data Structure and Algorithm using C

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is an example of a non-linear data structure? a. Array b. Stack c. LinkedList d. Tree	(L1, CO1)	1
2	Which of the following is NOT a type of data structure? a. Array b. Stack c. Hash Table d. Compiler	(L2, CO1)	1
3	Which data structure is best for implementing a LIFO (Last In, First Out) system? a. Stack b. Queue c. Tree d. LinkedList	(L3, CO2)	1
4	What is the time complexity of searching in a balanced binary search tree? a. $O(1)$ b. $O(n)$ c. $O(\log n)$ d. $O(n \log n)$	(L2, CO4)	1
5	In which type of linked list does the last node point to the first node? a. Singly Linked List b. Doubly Linked List c. Circular Linked List d. None of the above	(L1, CO1)	1
6	What is the worst-case time complexity of linear search? a. $O(1)$ b. $O(n)$ c. $O(\log n)$ d. $O(n \log n)$	(L1, CO5)	1
7	What is the key difference between a stack and a queue? a. Stack is LIFO, Queue is FIFO b. Both are LIFO c. Stack is FIFO, Queue is LIFO d. Both are FIFO	(L2, CO3)	1
8	Which graph traversal algorithm uses a queue? a. Depth First Search b. Breadth First Search c. Inorder Traversal d. Postorder Traversal	(L3, CO4)	1
9	Which of the following operations is not performed by a binary search tree? a. Insert b. Delete c. Sort d. Reverse	(L4, CO5)	1
10	What is the time complexity of bubble sort in the worst case? a. $O(1)$ b. $O(n)$ c. $O(n^2)$ d. $O(n \log n)$	(L2, CO5)	1

11	Stack overflow occurs in the operations	(L3, CO2)	1
	a. Push b. Pop c. display d. StackEmpty		
12	Which type of queue overcomes the limitation of a linear queue by reusing vacant spaces?	(L2, CO3)	1
	a. Simple Queue b. Circular Queue c. Priority Queue d. None of the above		
Q. 2	Solve the following.		12
A)	Explain the following operations on singly linked lists a) insertion of node at beginning and middle b) Traversal	(L2, CO1)	6
B)	Write differences between array and linked list.	(L3, CO1)	6
Q. 3	Solve the following.		12
A)	Define queue? Explain enqueue and dequeue operation for simple queue?	(L2, CO3)	6
B)	Explain type of Queue? Compare Simple Queue and Circular queue? Advantages and Disadvantages of each?	(L4, CO3)	6
Q. 4	Solve Any Two of the following.		12
A)	Write an algorithm for binary search and explain its time complexity?	(L3, CO5)	6
B)	Write an algorithm for Insertion sort with an example.?	(L4, CO5)	6
C)	Write a pseudo code for Linear Search? Explain with example	(L3, CO5)	6
Q. 5	Solve Any Two of the following.		12
A)	Write an algorithm to traverse a binary tree in inorder, preorder, and postorder.	(L3, CO4)	6
B)	Construct the final binary search tree (BST) by inserting the following numbers in the given order: 40, 20, 50, 10, 30, 60, 25, 35, 55, 70, 5, 15, 65, 75, 45 1. Draw the final binary search tree. 2. Write the pre-order traversal of the tree.	(L3, CO4)	6
C)	Describe the adjacency matrix representation of graphs with an example.	(L2, CO4)	6
Q. 6	Solve Any Two of the following.		12
A)	Implement a stack using an array. Write a program to perform push and pop operations.	(L3, CO2)	6
B)	Evaluate the postfix expression 15 7 1 1 + - / 3 * 2 1 1 + + - where 15, 7, 1, and 3 are operands. Show step-by-step using a stack.	(L3, CO2)	6
C)	Write an algorithm to evaluate the postfix expression using stack	(L3, CO2)	6
*** End ***			